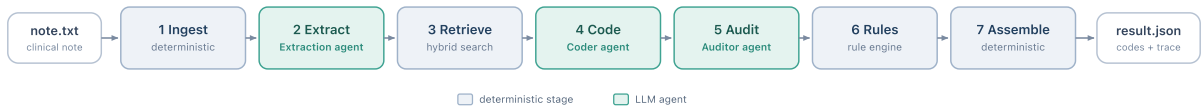


medcoder: Auditable Medical-Coding Pipeline

1 • Architecture & Data Flow

medcoder converts a free-text clinical note into billable codes (ICD-10-CM, the US clinical modification of ICD-10, for diagnoses; CPT for procedures) via a seven-stage pipeline (from `note.txt` to `result.json`) as shown in the diagram below. Three stages are LLM agents; the other four are plain deterministic code. Every stage is idempotent and can be rerun on its own, and every stage records its output to a trace file, so a reviewer can see exactly how each code was reached.



Pipeline at a glance

#	Stage	Runs on	What it does	Output
1	Ingest	deterministic	normalises text, splits SOAP sections, windows long notes, keeps character offsets	clean, sectioned note
2	Extract	Extraction agent	pulls clinical facts (verbatim span, normalised term, present/absent/possible, dx/px)	list of facts
3	Retrieve	deterministic	hybrid search returns a shortlist of real codes per fact (§2)	candidate codes
4	Code	Coder agent	picks 1 to N codes per fact, only from the shortlist, with a quoted rationale	assigned codes
5	Audit	Auditor agent	a second, independent model re-checks each (evidence, code) pair	agree / disagree notes
6	Rules	deterministic	ICD-10-CM guideline checks (conflicts, missing detail, format)	typed warnings
7	Assemble	deterministic	blends the signals into a confidence tier and writes the result plus its trace	result.json

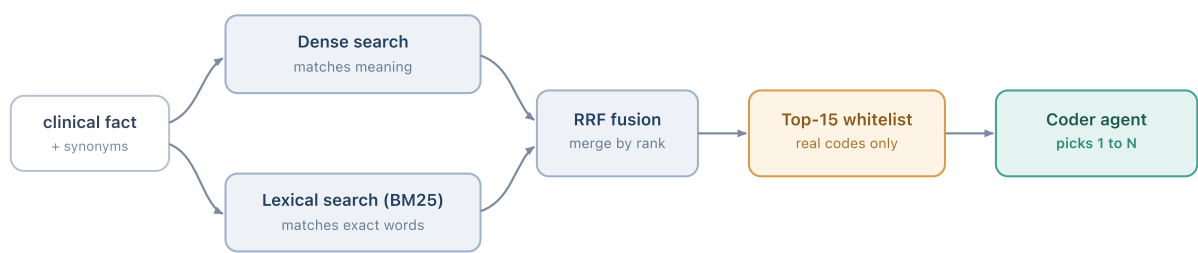
- 1. Ingest.** Normalises whitespace, segments SOAP sections, windows multi-page notes with overlap, and preserves global character offsets so every later span maps back to the source. It also detects the encounter type (inpatient vs outpatient), which governs whether “probable” or “suspected” diagnoses may be coded at all (ICD-10-CM Official Guidelines, Section IV.H [1]).
- 2. Extract (Extraction agent).** Returns a list of facts: the verbatim span, a normalised clinical term, the assertion status (present, absent, possible, hypothetical, family, historical), and the kind (diagnosis, procedure, symptom). A deterministic NegEx/ConText-style backstop overrules clear polarity mistakes, so “denies chest pain” is never coded as active chest pain.
- 3. Retrieve.** Hybrid search per fact, described in §2.
- 4. Code (Coder agent).** Picks 1 to N codes for each fact, only from the retrieved shortlist, with a stated confidence and a rationale that quotes the evidence span.
- 5. Audit (Auditor agent).** A separate model re-reads each (evidence, code) pair against the cited text and the code description, and returns agree or disagree with a short note. To control cost it is selective: only procedures and low-confidence diagnoses go through the auditor.
- 6. Rules.** Deterministic ICD-10-CM checks: Excludes1 conflicts (codes the guidelines say cannot be reported together), an unspecified code used where a specific one exists, a missing 7th character on injury or pregnancy codes, evidence anchoring, and basic format validity. Each issue becomes a typed warning (missing information, ambiguity, or conflict).
- 7. Assemble.** Blends three confidence signals (§3) into a single high, medium, or low tier and writes a self-contained outputs/<doc_id>/ folder. Each run writes one result file (`result.json` by default, `result.md` for a Markdown review sheet, or `result.annotated.md` for a copy of the note with codes spliced in at their evidence spans) together with a `trace.json` that records every stage’s output, so a reviewer can reconstruct how each code was reached.

Worked example. The phrase “Type 2 diabetes with diabetic polyneuropathy” becomes one fact, retrieves a 15-code shortlist, and the coder selects E11.42 with that phrase quoted as evidence; the auditor confirms it and the rule engine checks the code is billable.

2 • Code retrieval / Filtering strategy

Design: the coder is constrained to select only from a retrieved candidate set and can never free-generate a code. For each extracted fact, a short list of real catalog codes is retrieved; the coder may choose only from that list, and any code not on it is discarded during validation. Invalid codes are therefore impossible by construction, which matters because models left to free-generate ICD-10 produce up to about 35% of codes that are non-billable or do not exist [2].

How the shortlist is built. For each fact, two complementary searches are run over the code descriptions and combined:



- **Dense search** matches by meaning (semantic similarity via cosine), so “DM type 2” still finds “type 2 diabetes mellitus”. Sentence-transformer embeddings are queried over a FAISS index, returning the top 50.
- **Lexical search (BM25)** matches the exact wording, which matters when a phrase like “essential hypertension” must be read word for word. It also returns the top 50.
- **Reciprocal Rank Fusion** merges the two lists by rank rather than score [3], so two very different scoring scales need no calibration. The top 15 fused results become the shortlist.

Query expansion. Before searching, each fact carries a few synonyms the extraction agent already produced (for example “MI” expands to “myocardial infarction”). All are searched and the best result per code is kept, which widens recall without weakening the shortlist constraint.

Data: the real CDC ICD-10-CM FY2027 catalog (about 74,879 codes, US public domain) is bundled, because a toy catalog would dodge the large-vocabulary challenge the exercise is really testing. CPT is AMA-copyrighted with no free tier, so a clearly marked synthetic CPT-shaped catalog is shipped, and a licensed real-CPT catalog can be substituted through configuration without code changes.

3 · LLM usage & Prompting approach

Three role-specialised agents mirror the real coder-then-QA workflow. All of them go through one LiteLLM gateway, so providers and per-agent models swap with an environment variable.

Agent	Human analog	Model (default)	Key constraint
Extraction	clinical scribe	openai/gpt-5.4-mini	reason first, then emit schema-validated JSON
Coder	medical coder	openai/gpt-5.4-mini	may only choose codes from the shortlist
Auditor	QA reviewer	anthropic/claude-haiku-4-5-20251001	a different model family by default

Why a different family for the auditor. Using a different model family for the check reduces correlated errors and self-preference bias, where a model rubber-stamps its own answer [4]; a coder-plus-independent-auditor design also gives the best precision and recall in recent clinical-coding work [5]. A same-model fallback is supported but flagged as weaker.

Cost discipline. The extraction call does triple duty in one shot: it returns the facts, a note-level encounter type (replacing brittle keyword counting, \$1), and the per-fact synonyms that feed retrieval (\$2). Verification is selective (only procedures and low-confidence diagnoses) and calls are batched (one extract, one code, one audit per note).

Prompting choices.

- *Reason first, then format.* Forcing JSON-only output degrades reasoning quality [6], so prompts let the model think internally and then emit one JSON object validated against a schema. On a validation failure we retry once with the error appended, which is almost always enough.
- *Versioned prompts.* Prompt files are part of the config hash, so any prompt change shows up in the audit log.
- *Bounded, pinned sampling.* Models are pinned by dated ID. Temperature 0 is used where the provider honours it; GPT-5 reasoning models reject it, so determinism instead rests on structured outputs and a low `reasoning_effort` setting that is itself part of the config hash.

Confidence is not the raw LLM number. A model’s stated confidence is systematically overconfident [7]. We instead blend three signals (the fused retrieval rank, the coder’s discounted confidence, and an auditor adjustment of +0.15 for agree or -0.30 for disagree) and bin the result into high, medium, or low tiers using gold-tuned thresholds. Formal calibration is wired as an extension but needs a larger labelled set than the demo has.

4 · Key decisions & trade-offs

Decision	Why	Trade-off accepted
Retrieve, then constrain (shortlist, not free generation)	Removes invalid codes and turns a 75k-way generation problem into a 15-way choice [8]	Bounded by retriever recall, now measured as a separate stage in the eval
Coder plus independent auditor	Best precision and recall on clinical notes [5]; heterogeneous checkers cut correlated errors [4]	Extra LLM calls, reduced by selective and batched verification
Hybrid retrieval with RRF	Catches exact terms and paraphrases with no score tuning	Two indexes to build (about a minute on 75k codes)
Three fixed agents, no agent swarm	Speculative swarms add latency and silent errors without reliable accuracy gains [9]	Foregoes possible ensemble accuracy for predictability
Real ICD-10-CM plus synthetic CPT	Meets the large-vocabulary challenge and stays within AMA licensing	CPT demo runs on synthetic codes
LiteLLM gateway	One call for many providers, with a built-in mock for keyless tests	A thin extra dependency
Temperature 0 and pinned snapshots	Reproducibility is an explicit requirement	Foregoes a small accuracy gain from sampling
Deterministic rule engine	Auditable, exact, and free of LLM cost	Full guideline coverage needs the ICD-10 tabular data and CMS NCCI
CLI-first core	Matches the offline brief without over-building	No reviewer UI yet, though the JSON contract supports one

5 · Limitations & extensions

Limitations (stated honestly).

- *Assistive, not autonomous.* On the evaluation set the system reaches ICD-10 micro-F1 about 0.5; full-vocabulary ICD-10 coding is hard in general (reported state of the art is around micro-F1 0.54 [10]), so a human reviewer is required by design. The output (evidence spans, confidence tiers, editable reviewer fields) is shaped to make that review fast.
- *CPT is synthetic.* Real-CPT accuracy must be re-validated on a licensed catalog, which can be substituted through configuration without code changes.
- *General-purpose embedder.* A general-purpose embedder (`all-MiniLM-L6-v2`) is used by default because it is local, keyless, fast, and adequate for the short strings being embedded (code descriptions and extracted terms), where it works

well alongside BM25. For production clinical text, a domain-specific biomedical embedder (SapBERT [11] or PubMedBERT, trained for mention-to-concept linking) would improve recall; the embedder is pluggable via configuration.

- *Rule engine is a curated subset.* It covers format, billable status, a short Excludes1 list, 7th-character chapters, diagnosis-to-procedure linkage, and evidence anchoring; full coverage needs the ICD-10 tabular data and an NCCI refresh.
- *Confidence is threshold-tuned, not formally calibrated,* which needs a larger labelled set.
- *Evaluation is directional.* A small authored gold set (n=4) is used for directional signal only; the contribution is the metric methodology (micro precision, recall, and F1; exact-match ratio; hierarchical F1; and per-stage retrieval recall), not the absolute scores (ICD-10 micro-F1 about 0.5, CPT micro-F1 about 0.8). At n=4 a swing of about 0.05 between runs is normal sampling noise, so the committed metrics file is one representative run. A larger synthetic set would not add credibility; genuine validation needs a licensed labelled corpus such as MIMIC-IV. Precision (about 0.4) is over-coding-bound (the coder emits more codes than the gold labels), so the main lever is a more selective coder, not retrieval.

Extensions (designed for, not built). Several extensions are planned but not implemented: a Postgres-backed retriever (semantic, full-text, and fuzzy search in one datastore via pgvector, tsvector, and pg_trgm); domain-specific biomedical embeddings with a SNOMED-to-ICD crosswalk; a fuller rule engine (the complete ICD-10 tabular guidelines plus CMS NCCI); a FastAPI service with a reviewer UI; self-consistency confidence; a licensed real-CPT catalog; and LLM observability through LiteLLM callbacks. Because every stage is idempotent, the pipeline can also be orchestrated by a workflow engine such as Airflow or Celery.

References

- [1] ICD-10-CM Official Guidelines for Coding and Reporting, FY2026 (Section IV.H, uncertain diagnoses in the outpatient setting).
- [2] NEJM AI, 2024: large language models free-generate large fractions of invalid or non-billable ICD-10 codes.
- [3] Cormack, Clarke, and Buettcher, 2009: Reciprocal Rank Fusion outperforms individual rank-learning methods.
- [4] arXiv:2410.21819: heterogeneous verifiers reduce correlated errors and self-preference bias.
- [5] MDPI Informatics, 2026: a coder-plus-auditor LLM design gives the best precision and recall on MIMIC-IV coding.
- [6] arXiv:2408.02442: strict format-only (JSON) constraints degrade LLM reasoning quality.
- [7] Xiong et al., 2023: verbalised LLM confidence is systematically overconfident.
- [8] arXiv:2407.12849: retrieve-then-rerank constrained coding sharply improves valid-code accuracy.
- [9] MAST, NeurIPS 2025: multi-agent "swarm" systems add latency and silent errors without reliable accuracy gains.
- [10] RAG-Coding, 2026: state-of-the-art full-vocabulary ICD-10 coding peaks near micro-F1 0.54.
- [11] Liu et al., 2021: SapBERT, self-aligned biomedical embeddings for entity (mention-to-concept) linking.